QUANTUM CRYPTOGRAPHY (BB84 PROTOCOL) — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Quantum Cryptography (BB84 Protocol)

Question 1 10.0 points

Examine the Bloch sphere representation of the BB84 protocol quantum states in the diagram below. Describe the position of each of the four states ($|0\rangle$, $|1\rangle$, $|+\rangle$, $|-\rangle$) in terms of their polar and azimuthal angles on the Bloch sphere using rectilinear and diagonal bases.

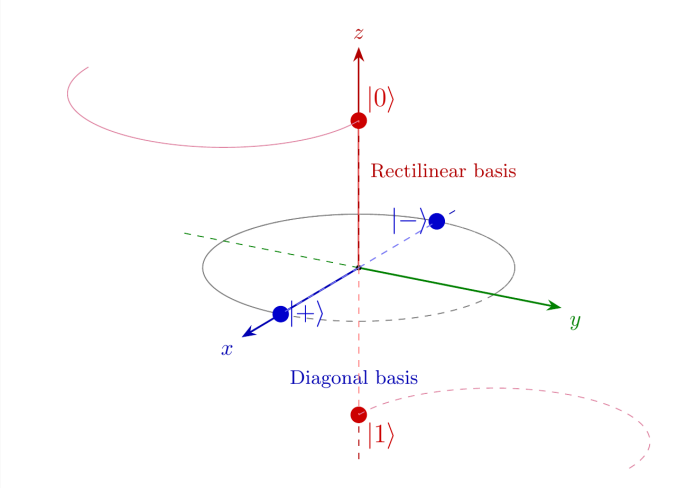

Fig. 1. A Bloch sphere with the quantum states |0⟩, |1⟩, |+⟩, and |-⟩ marked. The sphere should be oriented to show both the rectilinear basis (|0⟩ and |1⟩ along the z-axis) and the diagonal basis (|+⟩ and |-⟩ along the x-axis). The polar and azimuthal angles for each state should be indicated, with |0⟩ at the north pole, |1⟩ at the south pole, |+⟩ on the equator at 0 degrees azimuth, and |-⟩ on the equator at 180 degrees azimuth. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram fails to correctly label all four required. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram fails to show the quantum state. Regenerate using EXACTLY the values and sequences specified.

Question 2

10.0 points

Interpret the key exchange sequence in the diagram below for the BB84 protocol between Alice and Bob, including an intercept-resend attack by Eve. Explain how the protocol detects eavesdropping.

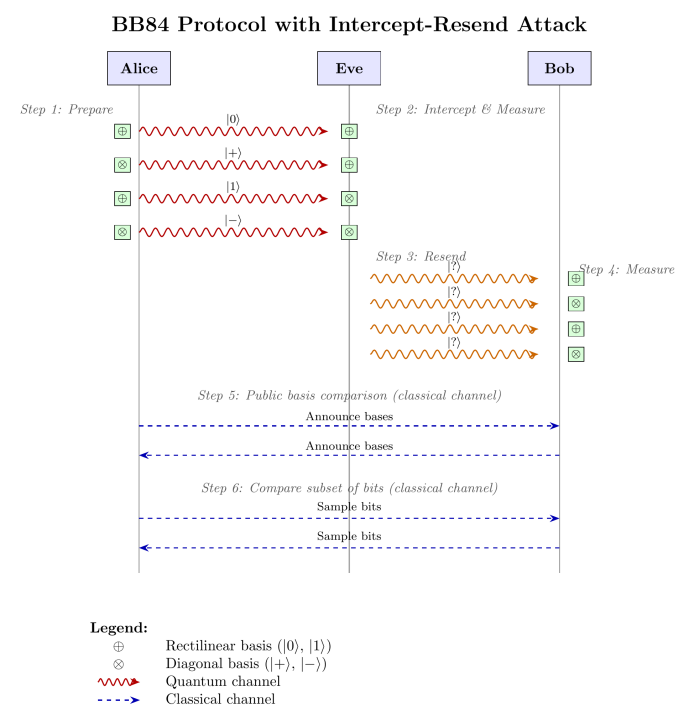

Fig. 2. A diagram illustrating the BB84 protocol key exchange sequence between Alice and Bob, including an intercept-resend attack by Eve. The diagram should show: 1. Alice preparing and sending qubits to Bob. 2. Eve intercepting and measuring the qubits, then resending them to Bob. 3. Bob receiving the qubits and measuring them. 4. Classical communication between Alice and Bob to compare bases and detect eavesdropping. Include labels for each step and indicate the quantum states and bases used.

Question 3

10.0 points

Examine the quantum key distribution channel in the diagram below, which illustrates the probability of eavesdropping detection versus sifted key error rate. Discuss the trade-off between security and error rate.

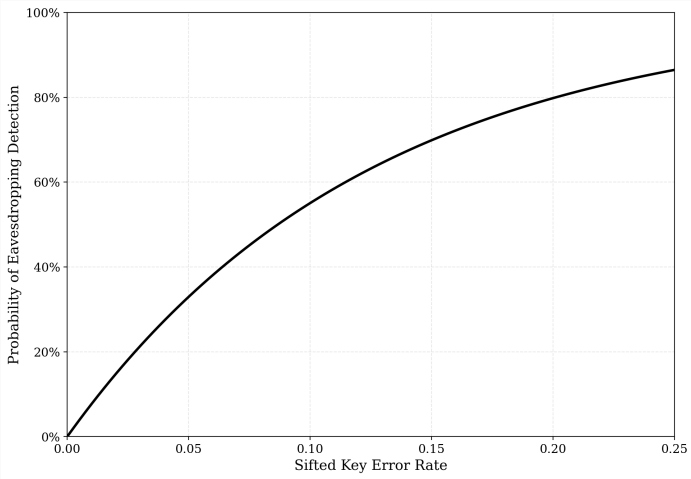

Fig. 3. A plot showing the probability of eavesdropping detection versus the sifted key error rate in a quantum key distribution channel. The x-axis represents the sifted key error rate, and the y-axis represents the probability of eavesdropping detection. The plot should illustrate how the probability of detecting eavesdropping changes as the error rate varies, highlighting the trade-off between security and error rate.

Question 4

10.0 points

Evaluate the quantum circuit shown in the diagram below, which demonstrates the no-cloning theorem. Explain how this circuit prevents the cloning of an arbitrary quantum state.

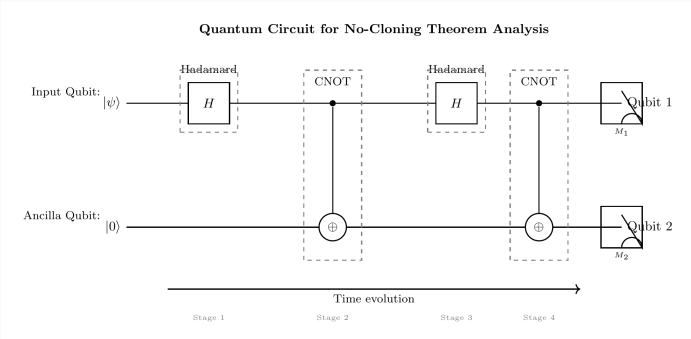

Fig. 4. A quantum circuit demonstrating the no-cloning theorem. The circuit includes a quantum state |ψ⟩ as input, a series of quantum gates such as Hadamard (H) and CNOT gates, and an auxiliary qubit initialized to |0⟩. The circuit should show how the gates are arranged to prevent cloning of the quantum state, with labeled qubits and gates. IMPORTANT correction needed: The diagram contains a label that explicitly states the. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Analyze the correlation curve between sifted key error rate and eavesdropping probability in the quantum key distribution session depicted in the diagram below. How can this curve guide secure communication practices?

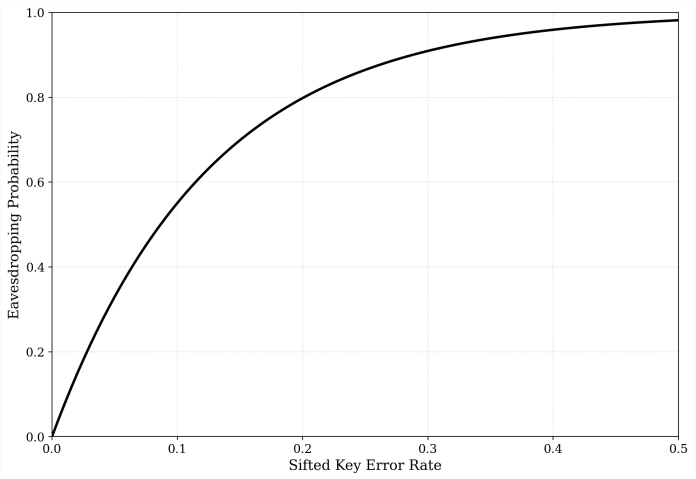

Fig. 5. A correlation curve showing the relationship between sifted key error rate and eavesdropping probability in a quantum key distribution session. The x-axis represents the sifted key error rate, and the y-axis represents the eavesdropping probability. The curve illustrates how changes in the error rate affect the probability of eavesdropping, which is crucial for understanding secure communication practices in quantum cryptography.